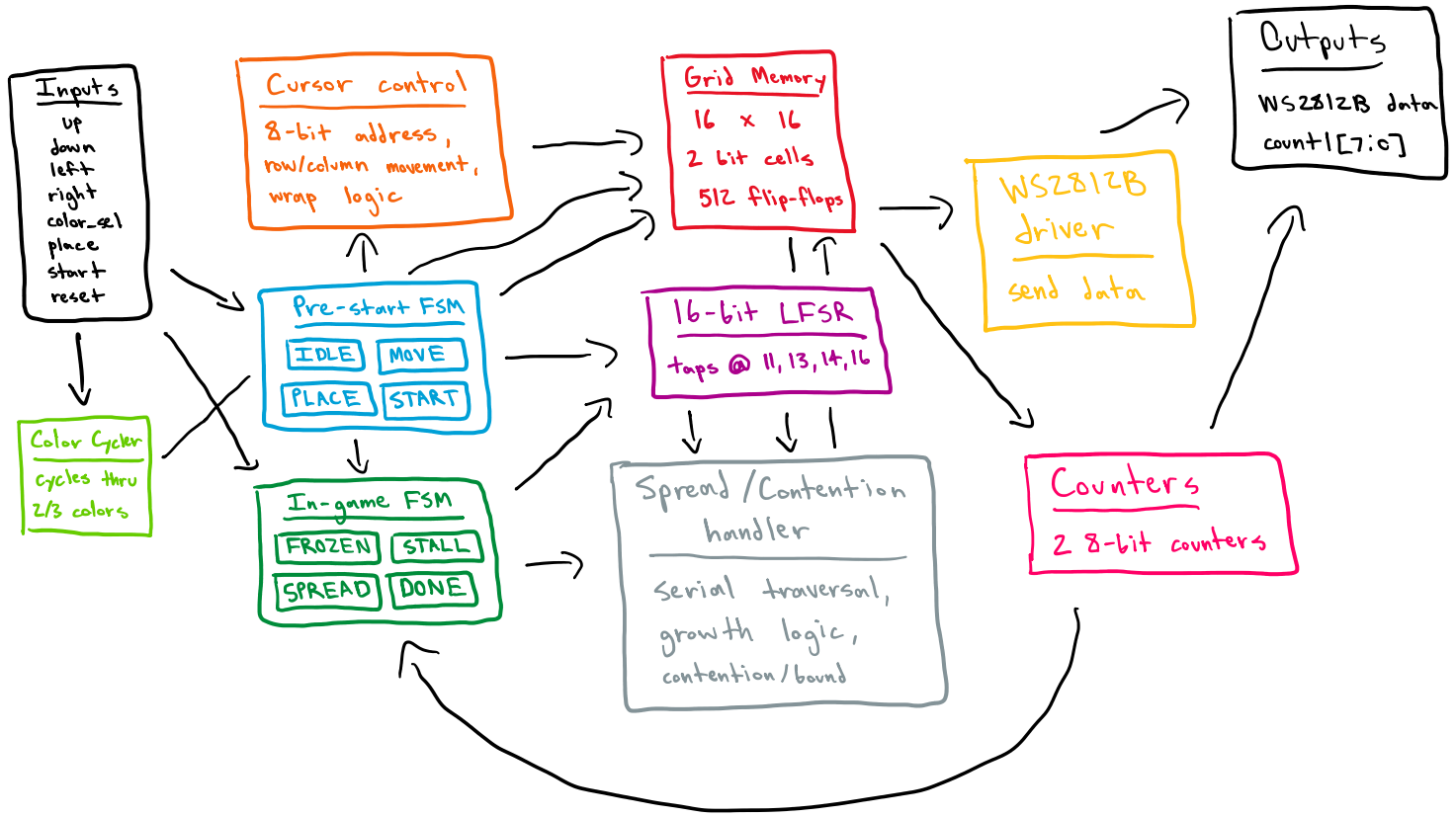
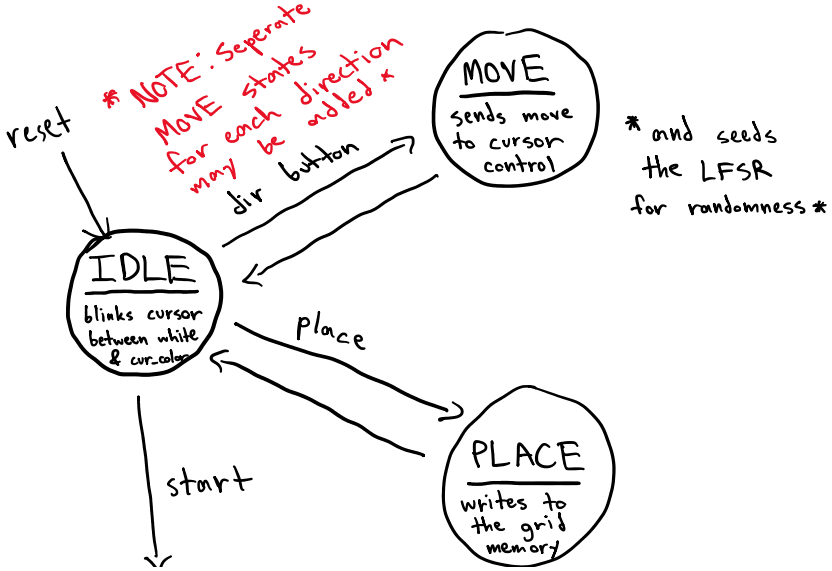
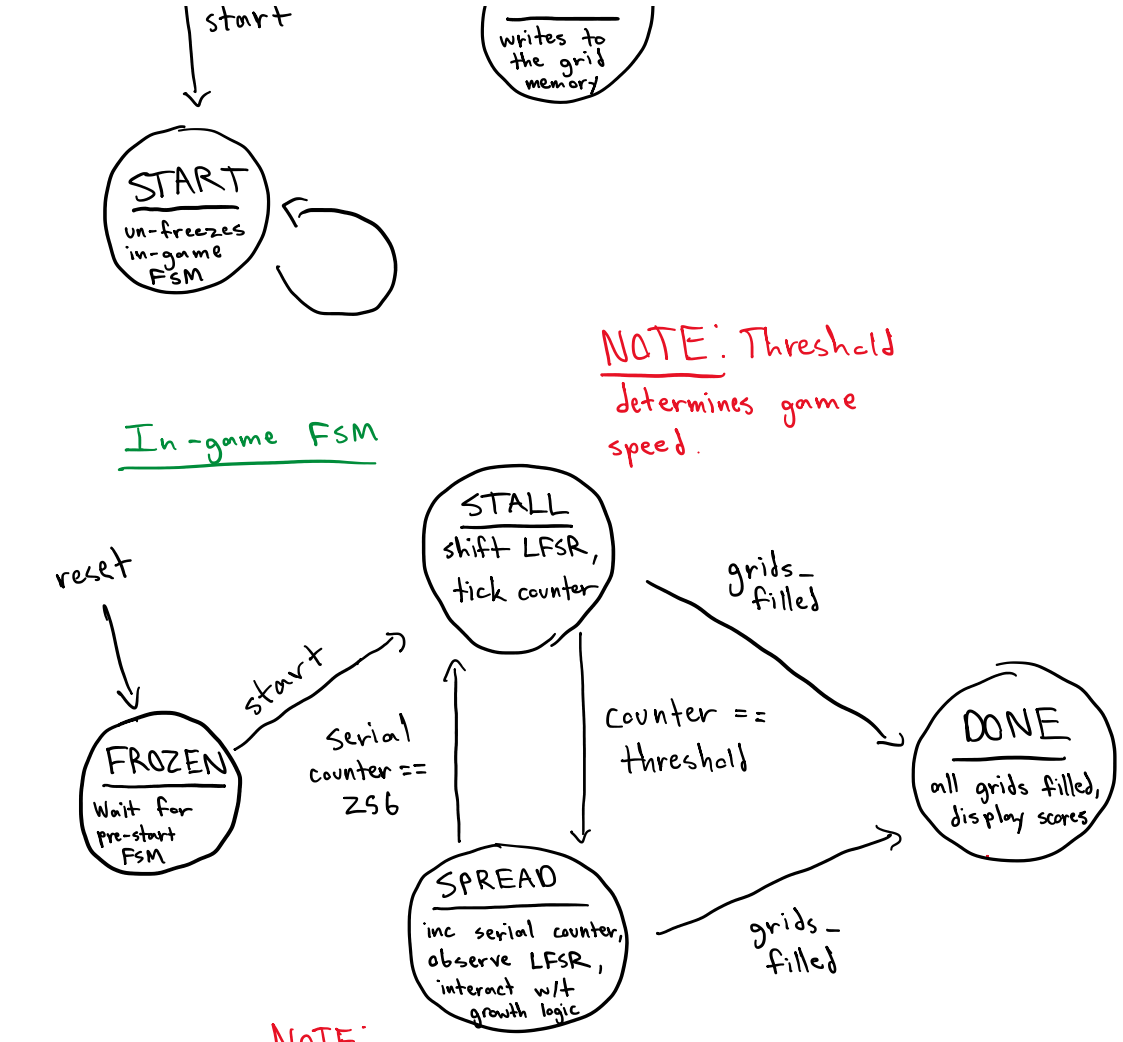
# Tile Growth Simulator Documentation




## Pre-start FSM

Note: This could be down w/t one state, but multiple states have easier readability *

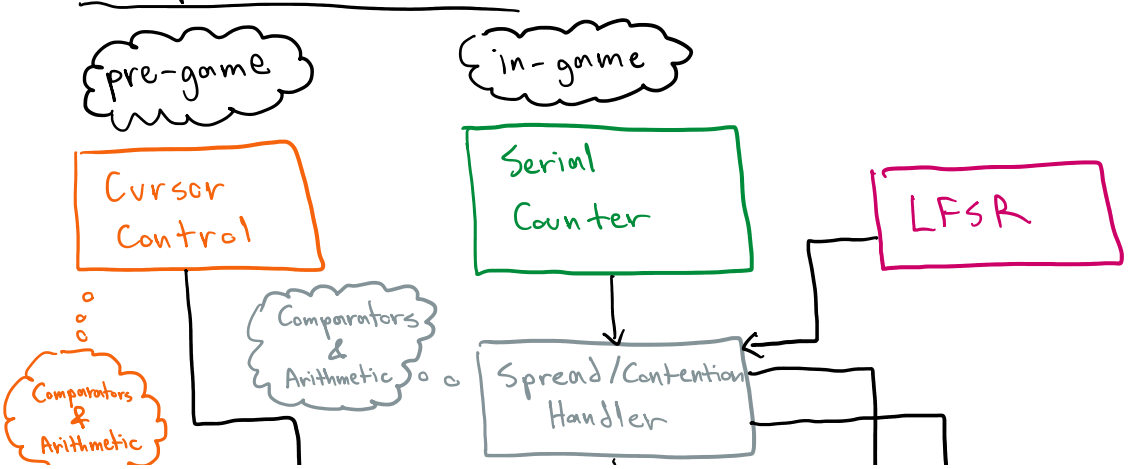


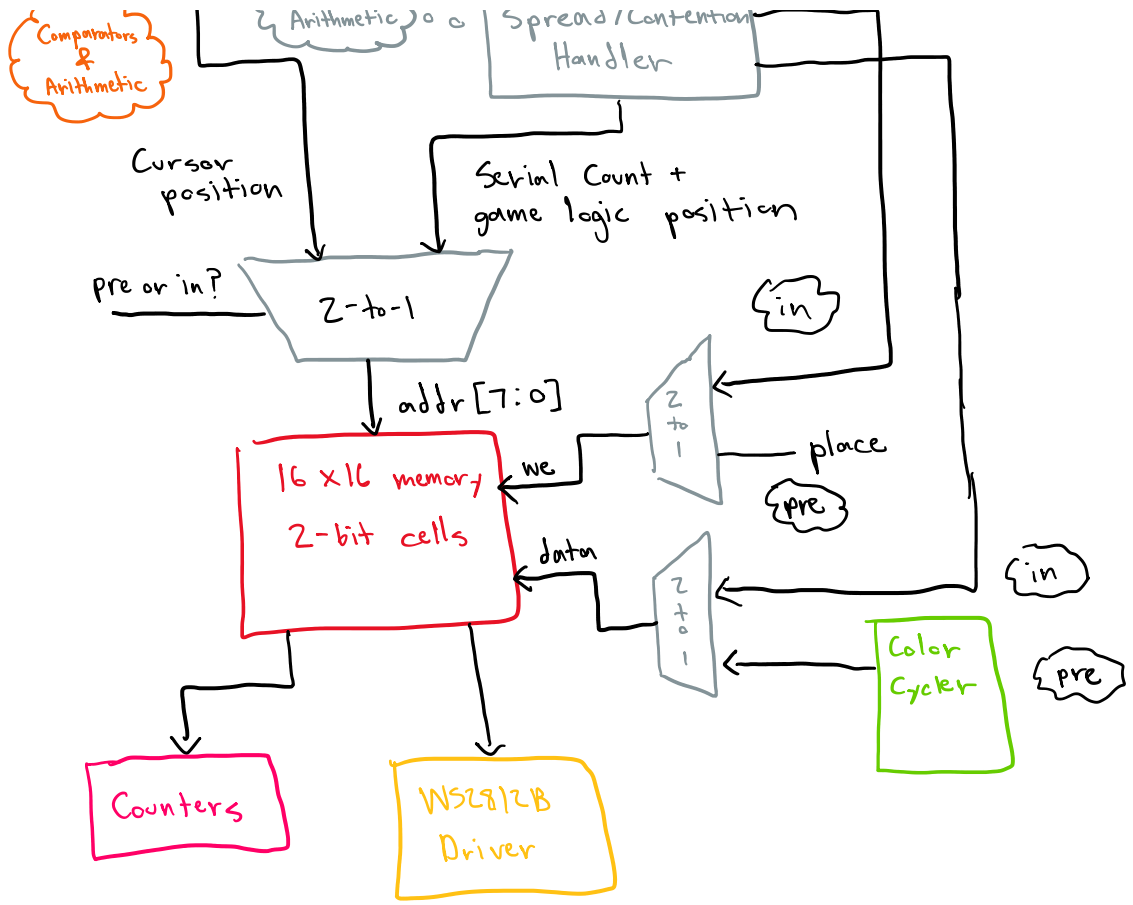



NOTE:

Each cell on the grid will have a unique 4-bits from the LFSR for growth determination and a unique 2-bits for direction determination

## Datapath schematic






## Testbench strategies

My general testbench strategy will be to stress test all of the modules in my design with tailored test vectors. Additionally, I will emulate my design on an FPGA and stress test it by pressing start multiple times, pressing multiple directional buttons at once, etc.

### Pre-start FSM

- Take all paths
- Press multiple dir buttons simultaneously
- Ensure start freezes the FSM

### In-game FSM

- Take all paths
- Ensure counts transition the states
- Ensure done stops all transitioning

### Cursor Control

- Reset places cursor @ (0,0)
- All four directions work
- Wrapper logic works

## Grid Memory

- Write to memory & ensure read matches
- Test all four colors at addresses
- Test edge addresses

## 16-bit LFSR

- Confirm LFSR value after specified seed
- Confirm that LFSR never becomes all 0's

## Spread/Contention Handler

- Block each direction in all configurations
- Block all four directions
- Confirm wrapper logic
- Confirm boundary logic

## WS2812B Driver

- Drive a vast array of test vectors
- Ensure protocol standards are met w/t assertions and waveform viewer.